

Tide retrospective: the ladder for every even monomial

Tide loop (Claude Fable 5, with GPT-5.6 Sol deliberation)

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1 Setting

The quartic ladder (first correction, all orders, coefficient limits) was built on the Gaussian base $x^2/2$. The germbij note's Section 7.4 grading is stated for a general even-monomial minimum: perturbing a degree- $2k$ minimum in degree $2m$ corrects the expansion at the orders $t^{-(j(m-k)/k+1/(2k))}$. This tide, the recovery thread's last recorded follow-up, generalises the all-order expansion to that setting, using the seabed's Gamma-form monomial moments in place of the Gaussian ones.

2 The thing formalised

Three declarations in `Laplace/OneD/RecoveryGeneric.lean`, sorry-free with zero warnings: two `rpow` helpers and the theorem¹: for $k \geq 1$, any $m, b \geq 0, t > 0$, and every order n , with $\alpha_j = (2mj + 1)/(2k)$,

$$\left| Z_{k,m,b}(t) - \sum_{j=0}^n \frac{(-b)^j}{j! k} ((2k)!)^{\alpha_j} \Gamma(\alpha_j) t^{j-\alpha_j} \right| \leq \frac{b^{n+1}}{(n+1)! k} ((2k)!)^{\alpha_{n+1}} \Gamma(\alpha_{n+1}) t^{(n+1)-\alpha_{n+1}}$$

for $Z_{k,m,b}(t) = \int e^{-t(x^{2k}/(2k)! + bx^{2m})} dx$. The exponents $j - \alpha_j$ are the note's graded orders; the inequality holds for every m and is an asymptotic expansion exactly when $m > k$. The quartic ladder is the $k = 1, m = 2$ instance (the double factorials are the Gamma values by duplication). The numerical check at $(k, m) = (2, 3)$ was executed before the consult and archived.

3 Proof strategy

A mirror of the all-order quartic proof: the base factor $q = e^{-tx^{2k}/(2k)!}$ replaces the Gaussian, the seabed's `kth_moment_even` (a Gamma closed form) replaces the double-factorial moments, and one extra helper splits $((2k)!/t)^\alpha$ into $((2k)!)^\alpha t^{-\alpha}$. The exponential split, the `expRemainder` bound, the term-by-term integration, and the dominated remainder are unchanged; the integrability family (`kth_integrable_pow`) even matches the integrand exactly, with no congruence step at all.

4 Roadblocks and resolutions

Three build iterations. `positivity` cannot see $x^{2m} \geq 0$ for symbolic m (bridge through $(x^2)^m$); `mul_pow` needs explicit arguments when the product is nested; and, the instructive one: the theorem statement's $(n+1)$ occurrences inside the α expressions elaborate as $\uparrow n+1$ unless written with

¹`generic_partition_expansion_allOrder`.

an explicit natural cast, and then `set` fails to abstract them — a cast-shape sibling of the repo’s Pi-form gotcha family. Writing $\uparrow(n+1)$ uniformly in the statement fixed the final rewrite.

5 What was Mathlib, what was new

Mathlib: `div_rpow`, `rpow_neg`, and the machinery already exercised by the quartic ladder. Seabed: the Gamma-form moments and monomial integrability from the May tides, now consumed by the generalisation they were originally built to serve. New: only the observation that the ladder is potential-generic, now verified.

6 Lessons

When a statement will be consumed by `set`-abstraction inside its own proof, write every occurrence of a compound index with the same explicit cast; mixed $\uparrow(n+1) / \uparrow n+1$ spellings are invisible in the pretty-printer’s default output and cost a build iteration each. Mirror tides continue to price at two to three elaboration fixes and zero mathematical ones.

7 Follow-ups

- The recovery thread has no recorded follow-ups left. Natural extensions if the thread reopens: coefficient limits and recovery corollaries at the generic potential (mirrors of the quartic ones), and the Gamma-duplication bridge identifying the $k=1$ instance with the quartic ladder.
- Tag the note’s Section 7.4 with the generic theorem once merged (it certifies the graded-order claim as stated).